

BYD is a leading EV maker in China that has been growing rapidly in scale, technology and overseas presence in recent years. But it's growing in a more uncertain environment marked by fierce domestic competition, trade frictions and world-wide volatility in energy and raw material markets, with U.S. tariff stimulus, European trade frictions and oil-price most of all amplifying Middle East war tension.

This report hones in on market risk, operational risk and country risk because these are most related to BYD's business profile. Market risk regards to the effect on profitability from price competition, commodity-input and exchange-rate fluctuations. Operational risk regards to the company's large-scale and fast-growing manufacturing, supply chains and delivery networks. Country risk becomes prominent in the context of overseas expansion that brings about trade frictions and regulatory pressure. It comes from a secondary risk of reputational risk in which a risk of quality issues and regulatory scrutiny or trade frictions involving the company intensify the core risks.

Market risk

Under this conceptual framework, market risk is measured by price competition risk, raw material price risk and foreign exchange risk as they are the primary external channels influencing BYD's finance metrics. Price competition entails cutbacks in selling price and margin, price volatility of raw material impacts costs and foreign exchange movements become more important as overseas exposure grows. These three dimensions together constitute a useful measure on market risk for BYD.

Table 1

Risk category	Why it is relevant to BYD	Main transmission channel	Focus of later analysis
Price competition risk	BYD operates in a highly competitive EV market, especially in China, where price cuts may pressure vehicle pricing.	Lower selling prices → lower revenue per vehicle → margin compression → weaker profitability	Vehicle sales growth, revenue growth, gross margin, net margin, average revenue per vehicle
Raw material price risk	BYD's production costs are sensitive to battery-related raw materials and broader input-price movements.	Higher input costs → higher cost of sales → lower gross margin → pressure on earnings	Cost of sales, gross margin trend, raw material price trend, cost sensitivity
Foreign exchange risk	Overseas expansion increases exposure to exchange-rate fluctuations and revenue translation effects.	FX fluctuations → translated revenue volatility / cost mismatch / exchange gains or losses → profit volatility	Overseas revenue ratio, FX gains/losses, foreign currency exposure, exchange-rate sensitivity

Price competition risk

Risk of price competition is a material market risk for BYD. In China’s highly intensified EV market, price cuts and changes in the product mix could help sales, but erode pricing power and profitability. An analysis of key 2023–2025 indicators is used to evaluate this risk.

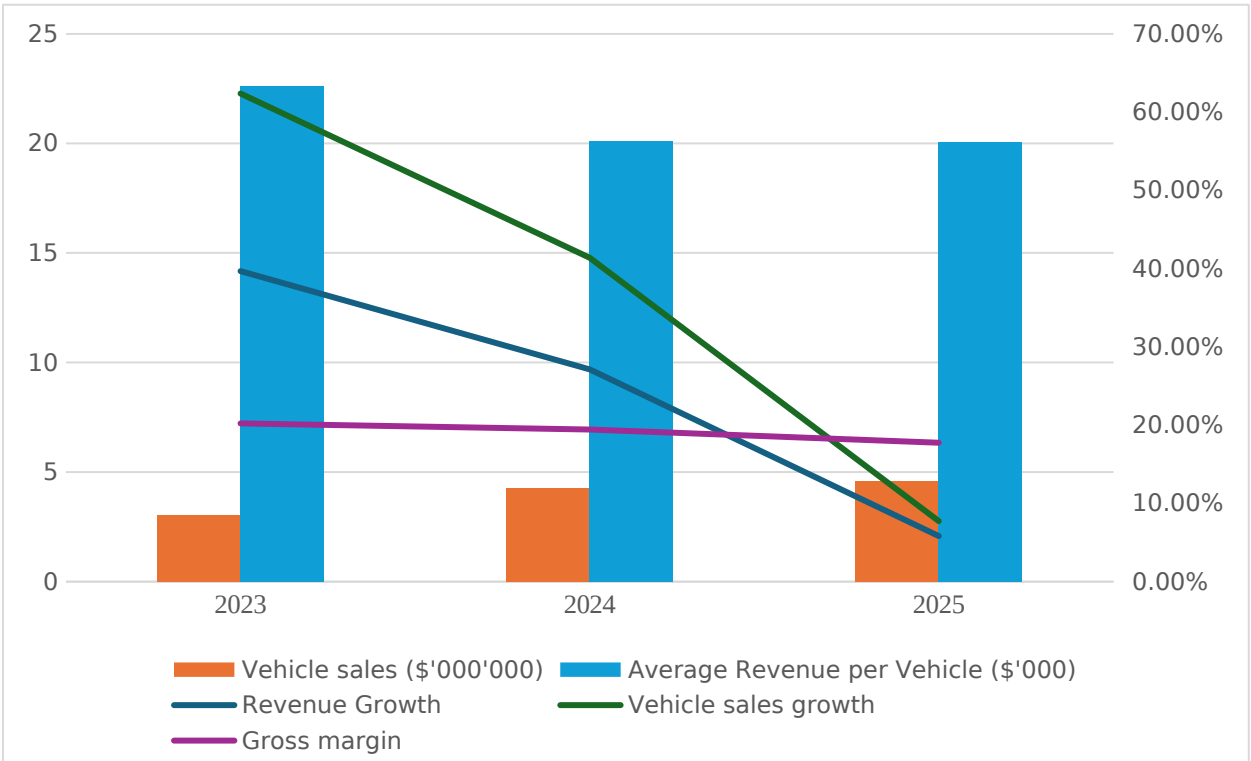


Figure 1
As shown in Figure 1, BYD’s vehicle sales continued to rise between 2023 and 2025, but growth slowed sharply. Revenue growth also decelerated, while average revenue per vehicle fell from 2.26 in 2023 to 2.01 in 2024 and 2025. Gross margin declined from 20.21% to 17.74% in 2025, indicating weaker pricing power and rising pressure on earnings quality.

Raw material price risk

This section assesses BYD’s raw material price risk using lithium carbonate as the main input-cost proxy. Lithium is selected because it is closely linked to battery-related costs and provides the most transparent basis for sensitivity analysis within the scope of this report.

Figure 2

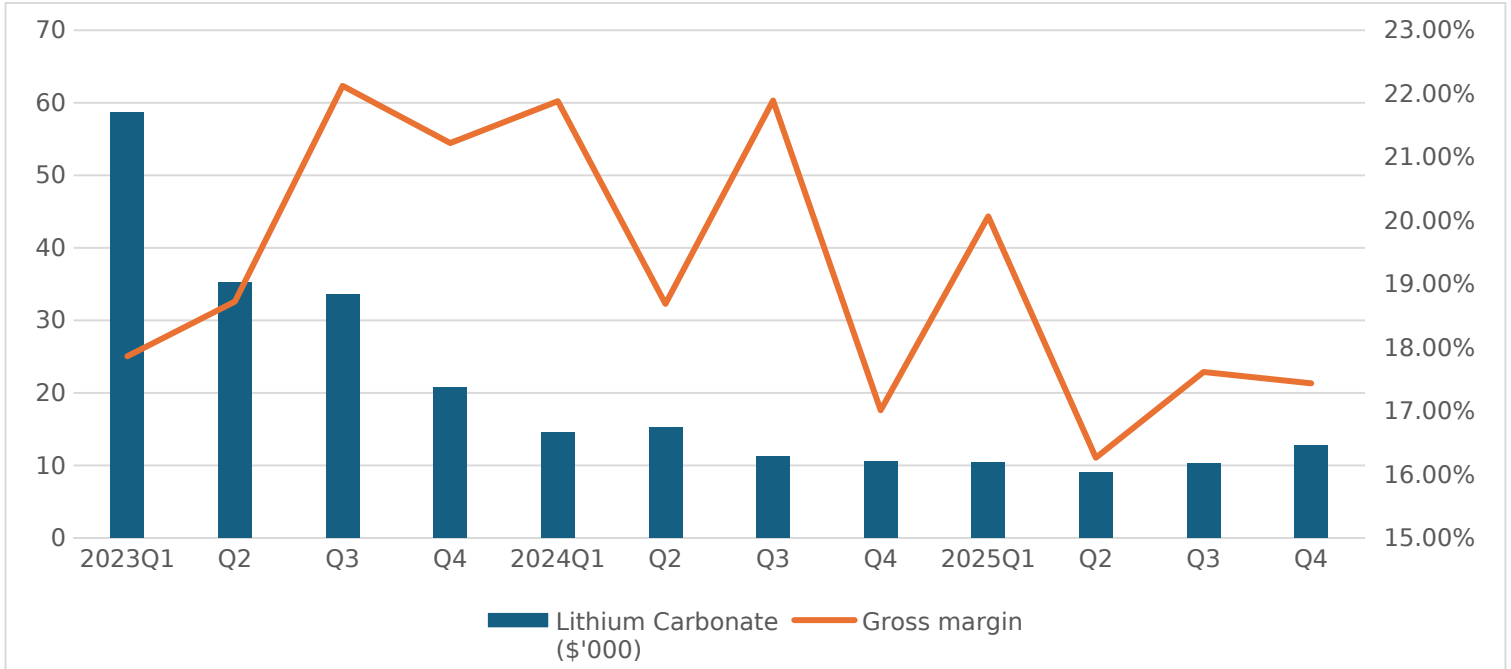


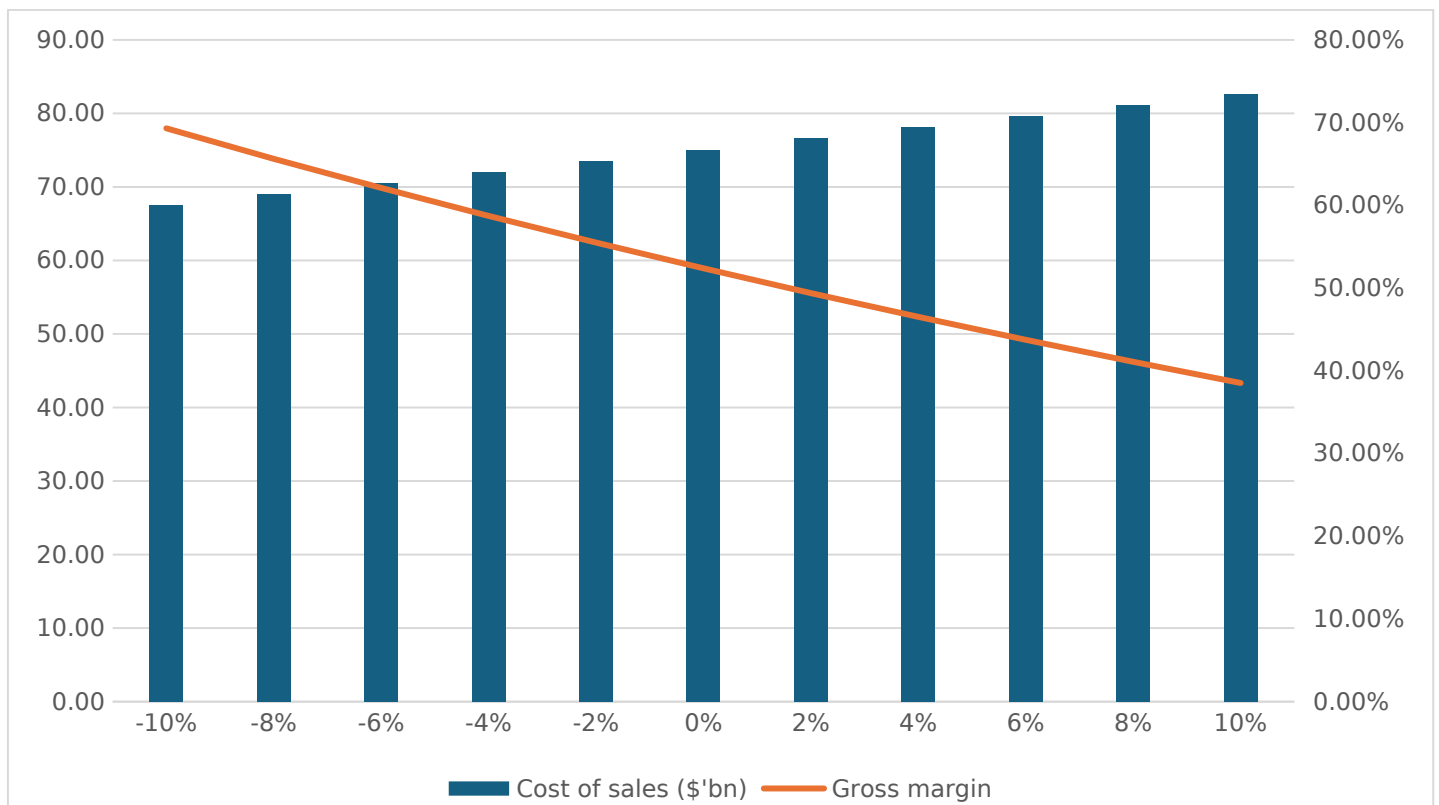
Figure 2 visualizes quarterly lithium carbonate prices and BYD gross margin against 2023–2025. Though this comparison is descriptive rather than causal, it indicates that lithium price variation can be aligned with profitability variation, thus supporting a cost-side sensitivity analysis.

To avoid assuming a 1:1 transmission between lithium price and cost of sales, the analysis uses a staged mechanism. Lithium carbonate first translates into battery cost and it is assumed that lithium price accounts for 15.4% of battery cost based on a CITIC Securities estimate. Then battery cost translates into vehicle cost, and battery packs are assumed to account for 35% of EV value with the IEA benchmark. Finally, vehicle cost is transmitted into BYD cost structure with vehicle cost accounting for 55.85% of BYD cost of sales in the 2025 annual report. Thus, the transmission is:

$$\Delta CoS = \cos_0 \times 35\% \times 15.4\% \times \Delta LC$$

On this basis, we conduct a sensitivity analysis of the price change of lithium carbonate on BYD's cost of sales and gross margin. As shown in Figure 3, an elevated price of lithium carbonate leads to higher cost of sales and lower gross margin, while negative shock will have opposite impact. It implies that the cost of lithium and other lithium related cost still pose a material risk for BYD.

Figure 3

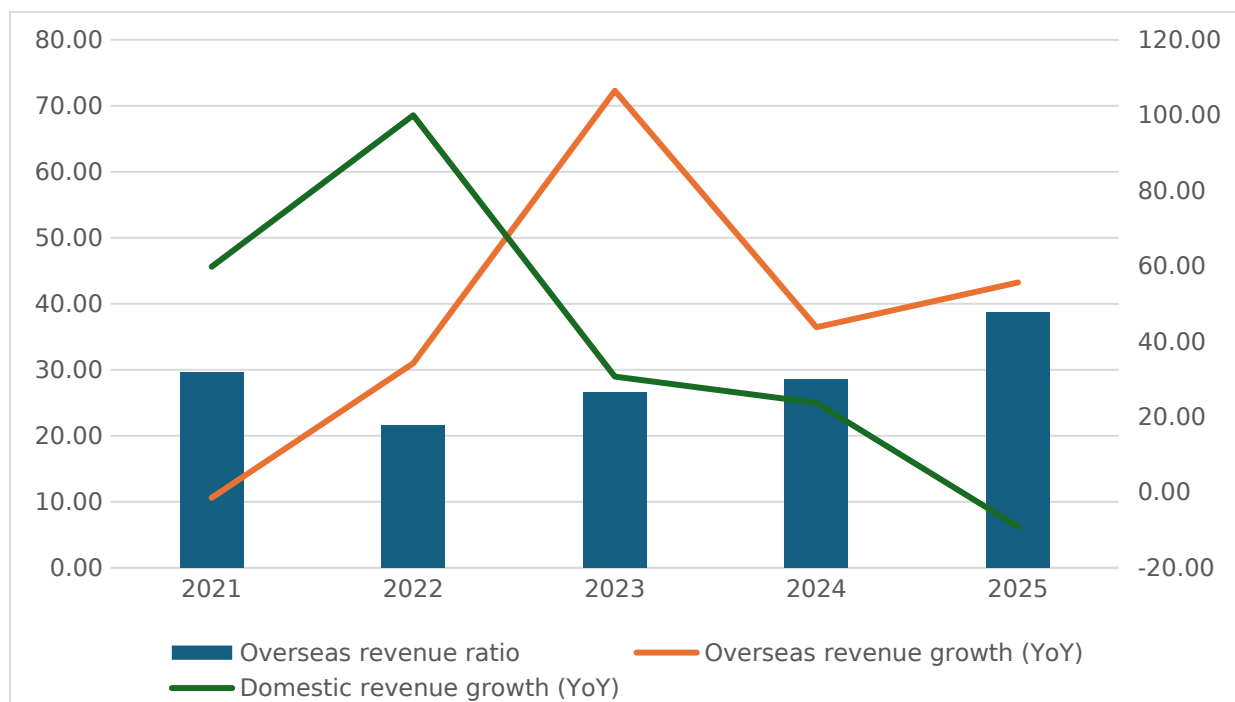


FX risk

Foreign exchange risk is becoming increasingly important as BYD expands abroad. With increasing foreign revenue, movements in exchange rates can affect the RMB value of sales and raise earnings volatility. This analysis therefore looks at whether BYD's increasing international exposure has made currency movements a more significant source of market risk.

As evidenced in Figure 4, domestic revenue growth peaked in 2022 and subsequently has declined to negative growth in 2025. In contrast, the share of overseas revenue has consistently increased since 2022, and overseas revenue growth has consistently exceeded domestic revenue growth, implying that BYD is increasingly reliant on foreign markets, and that exchange-rate movements are becoming more important to reported revenue and earnings stability.

Figure 4



While BYD's annual reports disclose overseas revenue but not regionally broken-down revenue figures, this report therefore does not assign foreign exchange exposure for BYD to any one non-RMB currency with any exactitude. Instead, USD/CNY is used to benchmark the overall external currency exposure.



Figure 5



Figure 6

Figures 5 and 6 indicate that since 2020, USD/CNY and EUR/CNY have largely tracked with the underlying 10-year bond yield differentials of the U.S./euro area against China. It implies that the long-term sovereign yield differentials serve as a key reference for the exchange-rate expectations. While not one-to-one over every short period, the trend overall shows that the long-term interest-rate differentials have been closely correlated with the movements of RMB against USD and EUR. Certainly, BYD does not disclose the currency composition of its overseas revenue, so this report adopts a very simplified two-factor FX sensitivity framework. An

exposure to the U.S. dollar is modelled in terms of the assumed USD share (w) of overseas revenue and the USD/CNY shock, with the sensitivity of the reported overseas revenue to be particularly measured as:

$$\text{Overseas Gross Profit}_1 = \text{Overseas Gross Profit}_0 \times (1 + w \times \Delta \text{USD} / \text{CNY})$$

Table 2 Sensitivity of Overseas Gross Profit (\$'bn)

$\Delta \text{USD/CNY}$	w	30%	40%	50%	60%	70%
-8%		8.40	8.33	8.26	8.19	8.12
-7%		8.42	8.36	8.30	8.24	8.18
-6%		8.45	8.40	8.35	8.29	8.24
-5%		8.47	8.43	8.39	8.35	8.30
-4%		8.50	8.47	8.43	8.40	8.36
-3%		8.53	8.50	8.47	8.45	8.42
-2%		8.55	8.53	8.52	8.50	8.48
-1%		8.58	8.57	8.56	8.55	8.54
0%		8.60	8.60	8.60	8.60	8.60
1%		8.63	8.64	8.65	8.65	8.66
2%		8.65	8.67	8.69	8.71	8.72
3%		8.68	8.71	8.73	8.76	8.78
4%		8.71	8.74	8.78	8.81	8.84
5%		8.73	8.78	8.82	8.86	8.90
6%		8.76	8.81	8.86	8.91	8.96
7%		8.78	8.84	8.90	8.96	9.02
8%		8.81	8.88	8.95	9.02	9.08

Table 2 implies that BYD's overseas profitability increases with the USD/CNY rate, and that the impact of exchange-rate changes increases with the assumed USD share of overseas revenue. Thus, BYD's reported overseas profitability is affected by exchange-rate changes, though the magnitude depends upon the assumed currency mix of overseas revenue.

Thus, the table indicates that FX risk is more likely to impact through overseas profitability translation than by large company-wide margin swings.

Conclusion

To hedge market risk, BYD could look at commercial and financial controls. Price competition risk could be partially hedged through product differentiation, flexibility in pricing and tighter cost control. Raw material price risk can be hedged through long term supply contracts, commodity diversification and selective commodity hedging where appropriate. Foreign exchange risk can be hedged with natural hedging, improved revenue and cost matching by

currency and forward contracts or other basic FX hedging instruments. All of these controls would hedge earnings volatility and improve BYD's earnings resilience.

Identify new risk

The second, and no less important, business risk for BYD is reputational risk. Reputational risk amplifies the market risk, operational risk and country risk identified in this report. BYD's rapid expansion overseas makes any product-quality or safety concerns, regulator investigations or trade controversies significant potentially eroding customer confidence and brand value. That is of particular concern to BYD because each bit of its success becomes contingent on foreign markets and acceptance by the consumer and the regulator. If reputational damage becomes real, the company may face weaker demand, slower overseas expansion, and stronger pressure on price and profitability. BYD should therefore improve product quality control, product transparency and the speed of response to customer complaints and regulatory issues. The company should also improve its compliance management overseas and strengthen its crisis-management system. These actions would all help to prevent reputational issues becoming financial or strategic problems.